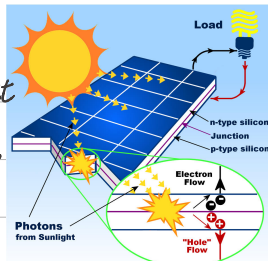


Q.1 Explain how crystalline Silicon converts sunlight into electricity in PV cells.

晶体的

utilizing the photovoltaics effect, where absorbed light energy generates charge carriers (electrons and holes), and the PN junction drives these carriers to create current and voltage across the external load



Solar insolation: Direct I_{bc} , Diffuse I_{dc} , Reflected I_{rc}

$$I_c = \overset{\text{most}}{I_{bc}} + I_{dc} + I_{rc}$$

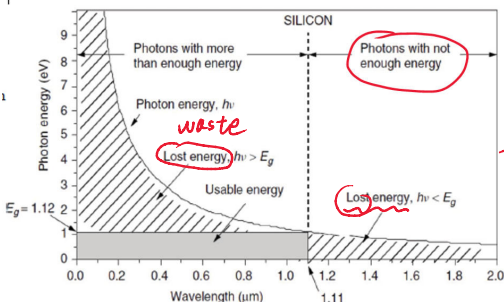
solar tracking systems increase annual power production by 20%~40%

Band gap energy for silicon is 1.12 eV $\Rightarrow$ 1.11 μm

all visible spectrum are strong enough to cause electrons to jump the band gap $\checkmark$

some infrared, all microwave, all radio waves $\times$

one photon can excite only one electron

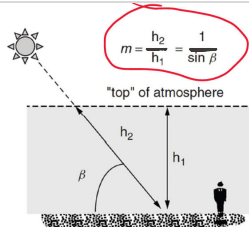


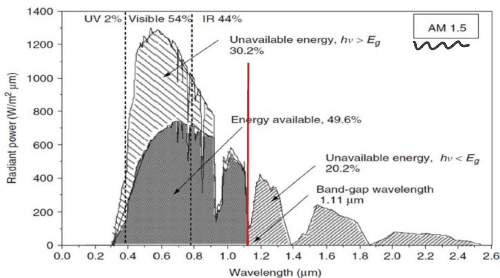
$\neq$ 100% efficiency

Air mass ratio (AM) (usually AM=1.5)

AM=1 the sun directly over head

AM=0 no atmosphere





The maximum possible fraction of the sun's energy that could be collected with a silicon cell is **49.6%**

band gap $\uparrow$, photons $\downarrow$, voltage $\uparrow$ current $\downarrow$

Q.2 What is the theoretical maximum conversion efficiency of Silicon based solar PV cells? Explain the reason behind this.

49.6%, due to losses caused by photons with insufficient or too much energy.

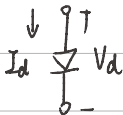
Q.3 What are the various losses in PV cells which reduce the conversion efficiency below the theoretical maximum efficiency?

- ① recombination of holes and electrons before contributing to current flow
- ② only about $\frac{1}{2}$ to $\frac{2}{3}$ of the full band-gap voltage across the terminals of the solar cell.
- ③ photons are not absorbed in the cell
- ④ internal resistance within cell, dissipating power

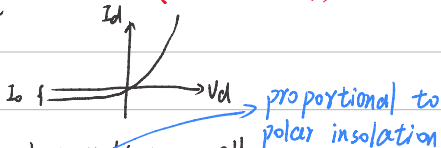
PN junction diode

$$I = I_{sc} - I_0 (e^{qV/KT} - 1)$$

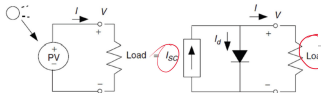
$$V_{oc} = \frac{KT}{q} \ln \left(\frac{I_{sc}}{I_0} + 1 \right)$$



$$I_d = I_0 (e^{38.9 V_d} - 1) \quad T = 25^\circ C$$



1. ideal equivalent circuit for a photovoltaic cell



$$I = I_{sc} - I_d = I_{sc} - I_0 (e^{38.9 V} - 1)$$

$$V_{oc} = 0.0257 \ln \left(\frac{I_{sc}}{I_0} + 1 \right)$$

Example

Area

Consider a 100-cm^2 photovoltaic cell with reverse saturation current $I_0 = 10^{-12} \text{ A/cm}^2$. In full sun, it produces a short-circuit current $I_{SC} = 40 \text{ mA/cm}^2$ at 25°C .

- Find the open-circuit voltage at full sun and again for 50% sunlight.
- Plot the results.

Solution

$$I_0 = 10^{-12} \times 100 = 1 \times 10^{-10} \text{ A}$$

At full sun:

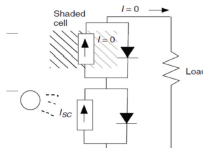
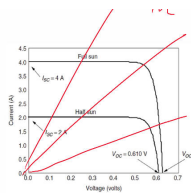
$$I_{SC} = 0.040 \times 100 = 4 \text{ A}$$

$$V_{OC} = 0.0257 \ln \left(\frac{I_{SC}}{I_0} + 1 \right) = 0.0257 \ln \left(\frac{4}{10^{-10}} + 1 \right) = 0.627 \text{ V}$$

At half sun:

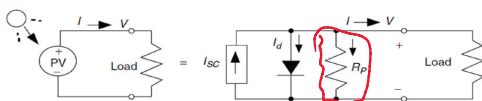
$$I_{SC} = \frac{4}{2} = 2 \text{ A}$$

$$V_{OC} = 0.0257 \ln \left(\frac{I_{SC}}{I_0} + 1 \right) = 0.0257 \ln \left(\frac{2}{10^{-10}} + 1 \right) = 0.610 \text{ V}$$



the simple equivalent circuit in series suggests no current can flow to load if any cell is in the dark

2. PV equivalent circuit with an added parallel resistance

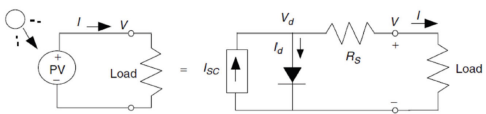


$$I = (I_{SC} - I_d) - \frac{V}{R_p}$$

for a cell to have losses of less than 1% due to parallel resistance

$$\frac{V_{OC}}{R_p} < 0.01 I_{SC} \Rightarrow R_p > \frac{100 V_{OC}}{I_{SC}}$$

3. PV equivalent circuit with an added series resistance



$$\begin{cases} I = I_{SC} - I_d = I_{SC} - I_0 (e^{qV_d/kT} - 1) \\ V_d = V + I \cdot R_s \end{cases}$$

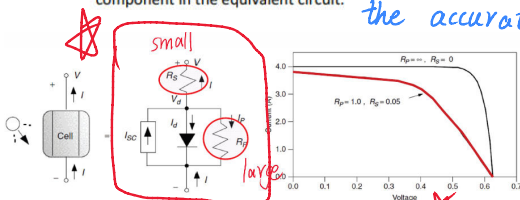
$$\Rightarrow I = I_{SC} - I_0 \left[\exp \left(\frac{q(V + I R_s)}{kT} \right) - 1 \right]$$

$$I_{SC} \cdot R_s < 0.01 V_{OC} \Rightarrow R_s < 0.01 \frac{V_{OC}}{I_{SC}}$$



Q.4 Draw the complete equivalent circuit of the solar PV cell. Explain the principle behind each component in the equivalent circuit.

the accurate equivalent circuit



$$I = I_{SC} - I_0 \left[\exp \left(\frac{q(V + I \cdot R_s)}{kT} \right) - 1 \right] - \left(\frac{V + I \cdot R_s}{R_p} \right)$$

$$I_{SC} = I + I_d + I_p$$

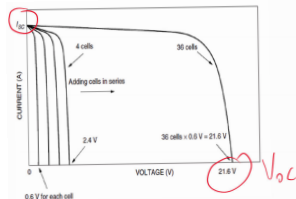
$$I = I_{SC} - I_0 \left[e^{\frac{38.9(V + I R_s)}{kT}} - 1 \right] - \frac{V + I R_s}{R_p}$$

- ★ Q.5 Find the open circuit voltage of solar PV cell with the following information given. Ambient temperature 40°C . Boltzmann constant $k = 1.381 \times 10^{-23}$, charge of electron $q = 1.602 \times 10^{-19}$ C. Short circuit current is 4A and the reverse saturation current is 10^{-10} A.

$$V_{oc} = \frac{kT}{q} \ln \left(\frac{I_{sc}}{I_0} + 1 \right) = \frac{1.381 \times 10^{-23} \times 313}{1.602 \times 10^{-19}} \ln \left(\frac{4}{10^{-10}} + 1 \right) \approx 0.65 \text{ V}$$

From cells to a module

A typical module has 36 cells, often designed as 12-V module



wired in series current — voltage ↑

$$V_{\text{module}} = n (V_d - I R_s)$$

Solution

Given $V_d = 0.5 \text{ V}$

$$I = I_{sc} - I_0 (e^{38.9 \times V_d} - 1) - \frac{V_d}{R_p} = I_{sc} - I_0 (e^{38.9 \times 0.5} - 1) - \frac{0.5}{6.6} = 3.16 \text{ A}$$

For a 36-cell module, voltage produced:

$$V_{\text{module}} = n(V_d - I R_s) = 36 \times (0.5 - 3.14 \times 0.005) = 17.43 \text{ V}$$

Power delivered by the module:

$$P = V \times I = 17.43 \times 3.16 = 55 \text{ W}$$

★ Example :

Voltage and current from a PV module Full sun 1000W/m²

A PV module is made up of 36 identical cells, all wired in series. With 1-sun insolation (1 kW/m²), each cell has short-circuit current $I_{sc} = 3.4 \text{ A}$ and at 25°C its reverse saturation current is $I_0 = 6 \times 10^{-10} \text{ A}$. Parallel resistance $R_p = 6.6 \Omega$ and series resistance $R_s = 0.005 \Omega$.

Find the voltage, current, and power delivered when the junction voltage of each cell is 0.50-V

- ★ Q.6 A solar PV module with 32 cells in series supplies 2A to a 5Ω load. The ambient temperature is 25°C . The parallel and series resistors in the PV cell equivalent circuit are 10Ω and 0.1Ω respectively. The reverse saturation current of the junction is 10^{-10} A . Determine the short circuit current of the cell.

$$I_0 = 10^{-10} \text{ A}, R_p = 10\Omega, R_s = 0.1\Omega, I = 2 \text{ A}, R = 5\Omega$$

$$V_d = \frac{2 \times 5}{32} \quad I = I_{sc} - I_0 \left[e^{38.9 (V + I R_s)} \right] - \frac{V + I R_s}{R_p}$$

- ★ Q.7 The loss due to series and parallel resistors in the equivalent circuit of the PV cell are less than 1 percent. Determine the limits of their values if the $V_{oc} = 0.6 \text{ V}$ and $I_{sc} = 4 \text{ A}$.

$$R_p > 100 \frac{V_{oc}}{I_{sc}}$$

$$R_s < 0.01 \frac{V_{oc}}{I_{sc}}$$

$$R_p > 15\Omega$$

$$R_s < 0.0015\Omega$$

- ★ Q.8 A PV module is made up of 40 identical cells, all wired in series. With 1-sun insolation (1 kW/m²), each cell has short-circuit current $I_{sc} = 4 \text{ A}$ and at 25°C its reverse saturation current is $I_0 = 10^{-10} \text{ A}$. Parallel resistance $R_p = 6 \Omega$ and series resistance $R_s = 0.01 \Omega$.

Find the voltage, current, and power delivered when the junction voltage of each cell is 0.60 V.

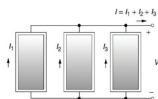
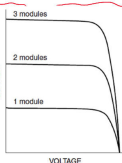
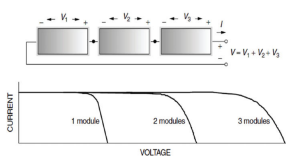
$$V_d = 0.6V$$

$$I = I_{sc} - I_0 [e^{\frac{38.9 V_d}{kT}} - 1] - \frac{V_d}{R_p} = 2.53 A$$

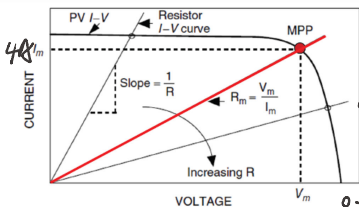
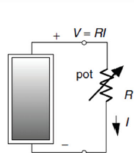
$$V_{module} = n (V_d - I R_s) = 40 \times (0.6 - 2.53 \times 0.01) = 22.99V$$

$$P = V_m I = 58.16 W$$

From modules to arrays



Maximum Power Point (MPP)



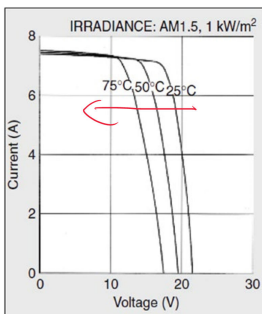
$$R_m = \frac{V_m}{I_m}$$

$$R = 1 \Omega$$

Q.9 A PV cell output can be approximated to be linear from the maximum power point till the open circuit voltage. MPP voltage and current are: $V_p = 0.5 V$ and $I_p = 4 A$. The open circuit voltage $V_{oc} = 0.6 V$. Determine operating point if a 1Ω resistor is connected to the PV cell.

$$R_p = \frac{0.5V}{4A} = 0.125 \Omega$$

Effect of I-V curves

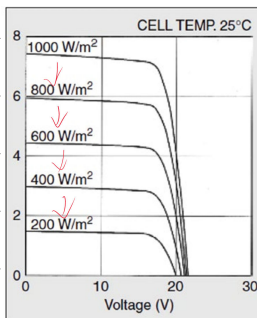


Temperature

V_{oc} drop 0.37%

I_{sc} 0.05%

MPP drop 0.5%
per degree C



Insolation

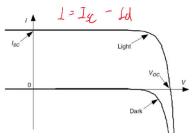
Q.10 What constitute standard test conditions (STC) for solar PV cells?

- Solar irradiance – 1 kW/m^2 (sun)
- Air mass ratio of 1.5 AM
- Cell temperature 25°C

Q.11 How does solar insolation affect the short circuit current and open circuit voltage of the solar PV cells?

$$I = I_{SC} - I_0(e^{38.9 V} - 1)$$

$$V_{OC} = 0.0257 \ln \left(\frac{I_{SC}}{I_0} + 1 \right)$$



- In both these equations, short circuit current is directly proportional to solar insolation, hence we can now plot current-voltage curves for varying sunlight.
- The dark (no sunlight) curve is just the diode curve turned upside-down. The light (illuminated cell) curve is the dark curve plus I_{SC}

Effect of Insolation

- As insolation drops, I_{SC} drops proportionately
- V_{OC} drops very slightly

Q.12 Explain the effect of partial shading in solar PV modules? How is the effect mitigated in practical circuits?

$I_d = 0$ Voltage is reduced by $I R_p$

Effect of shading..

Output voltage of the module becomes

$$V_{SH} = V_{n-1} - I(R_p + R_s)$$

For the remaining cells, $V_{n-1} = \left(\frac{n-1}{n} \right) V$

Substituting, $V_{SH} = \left(\frac{n-1}{n} \right) V - I(R_p + R_s)$

The drop in voltage due to shading of one cell:

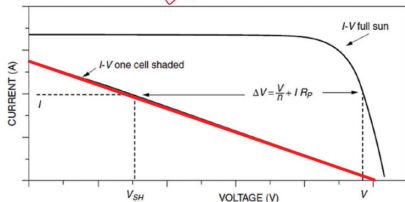
$$\Delta V = V - V_{SH} = V - \left(\frac{n-1}{n} \right) V + I(R_p + R_s)$$

$$\Delta V = \frac{V}{n} + I(R_p + R_s)$$

Since R_s is very small,

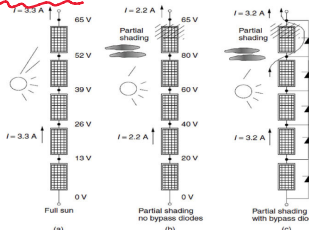
$$\Delta V \cong \frac{V}{n} + I R_p$$

$$\Delta V \cong \frac{V}{n} + I R_p$$



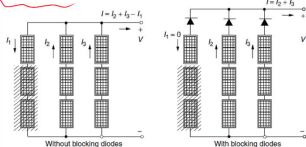
① Bypass Diode to mitigate effect of shading..

② Blocking Diodes



With bypass diodes, current is diverted around the shaded module.

- For strings of modules connected in parallel, if one of the strings is not performing well, it tends to draw current
- Blocking diodes can prevent the reverse current drawn by shaded string



Solution

The drop in module voltage will be $\Delta V = \frac{V}{n} + I R_p$

$$= \frac{19.41}{36} + 2.14 \times 6.6 = 14.66 \text{ V}$$

The new output voltage will be $19.41 - 14.66 = 4.75 \text{ V}$

Power delivered by the module with one cell shaded would be

$$P_{\text{module}} = VI = 4.75 \text{ V} \times 2.14 \text{ A} = 10.1 \text{ W}$$

For comparison, in full sun the module was producing 41.5 W . All of that 2.14 A of current goes through the parallel plus series resistance (0.005Ω) of the shaded cell, so the drop across the shaded cell will be

$$V_c = I(R_p + R_s) = 2.14(6.6 + 0.005) = 14.14 \text{ V}$$

Example

The 36-cell PV module described in previous example had a parallel resistance per cell of $R_p = 6.6 \Omega$. In full sun and at current $I = 2.14 \text{ A}$ the output voltage was found to be $V = 19.41 \text{ V}$. $R_s \approx 0.005 \Omega$

If one cell is shaded and this current somehow stays the same,

- What would be the new module output voltage and power?
- What would be the voltage drop across the shaded cell

Q.13 A 40-cell PV module has a parallel resistance per cell of $R_p = 5 \Omega$ where as the series resistance is $R_s = 0.01 \Omega$. In full sun and at current $I = 3 \text{ A}$ the output voltage was found there to be $V = 24 \text{ V}$. If one cell is shaded and the current somehow stays the same, then:

- What would be the new module output voltage and power?
- What would be the voltage drop across the shaded cell?
- How much power would be dissipated in the shaded cell?

$$(a) \Delta V = \frac{V}{n} + I R_p = \frac{24}{40} + 3 \times 5 = 15.6 \text{ V}$$

$$V_o = V - \Delta V = 24 - 15.6 = 8.4 \text{ V}$$

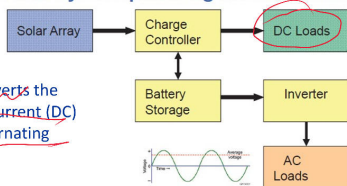
$$P = V_o I = 8.4 \times 3 = 25.2 \text{ W}$$

$$(b) V_c = I (R_p + R_s) = 3 \times (5 + 0.01) = 15.03 \text{ V}$$

$$(c) P_c = V_c \cdot I = 15.03 \times 3 = 45.09 \text{ W}$$

Q.14 Using block diagram show the components of a stand-alone Solar PV system, for a remote home. Describe the function of various components.

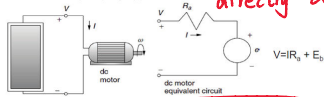
Typical stand-alone system powering DC and AC loads:



- The inverter converts the system's direct current (DC) electricity to alternating current (AC)
- Charge controller regulates the battery charging/discharging

Load-coupled PV Systems – DC Motor for pumps

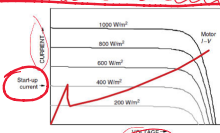
- Typical motor for water pumping applications



The motor will not start pumping until insolation reaches 400 W/m^2

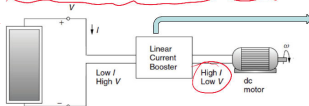
The system has to be carefully designed!

Insufficient insolation in the mornings results in lower efficiency

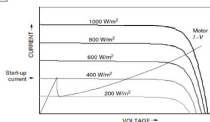


VOLTAGE

Linear current booster (LCB)



A linear current booster (LCB) increases current to help start or keep the motor running in low sunlight



VOLTAGE

Q.15 A remote home uses following appliances daily.

Appliance	Rating (W)	Daily usage duration (hours)
TV	100	10
Lights	50	12
Fan	100	20

Design the size of the solar PV panel and the battery capacity in Ah. If the 12V battery is supposed to store energy for about 3 days. Assume that the battery can be discharged up to 80% of its storage capacity. The location has 5 peak-sun hours on an average. Assume full-sun insolation is 1 kWh/m^2

$$\text{Energy (Wh/day)} = 100 \times 10 + 50 \times 12 + 100 \times 20 = 3600 \text{ Wh/day}$$

$$\text{all DC} = 3.6 \text{ kWh/day}$$

$$\text{Energy (whole)} = \frac{\text{Energy (kWh/day)} \times 3}{0.8} = 13.5 \text{ kWh}$$

$$\text{battery capacity} = \frac{13.5 \text{ kWh}}{12 \text{ V}} = 1125 \text{ Ah}$$

$$P_{ac} = \frac{3.6 \text{ kWh}}{5} = 0.72 \text{ kW}$$

assume $DC \rightarrow AC$ 75%

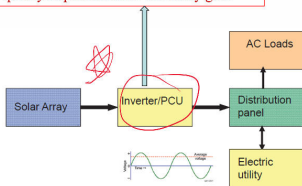
Q.17 Using detailed block diagram, show the components of a grid-connected solar PV system. Describe the function of each component.

Grid-connected PV Systems...

The components of a grid-connected PV system include:

- PV modules/arrays
- An inverter and power conditioning unit (PCU)
- A safety device to power down at failures in the grid and an electricity meter
- Designed to operate in parallel with, and interconnected, with the electric utility grid

The primary component in grid-connected systems is the inverter/PCU which converts the DC power into AC power consistent with the voltage and power quality requirements of the utility grid



$$P_{dc} = 1 \text{ kW/m}^2 \text{ insolation} \cdot A (\text{cm}^2) \cdot \eta$$

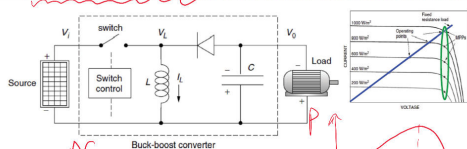
assume $\eta = 12.5\%$

$$A = \frac{0.96}{1 \text{ kW/m}^2 \times 0.125} = 7.68 \text{ m}^2$$

Q.18 Describe the operation of buck-boost converter used for MPPT operation in solar PV system.

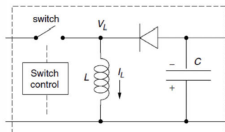
Maximum Power Point Trackers (MPPT)

- Standard part of most grid-connected systems
- Commonly use a buck-boost converter (also used in linear current boosters)
- Raises or lowers the voltage to the desired value needed by load

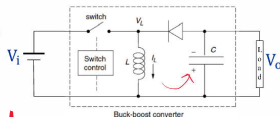


Buck-Boost Converter circuit

- A buck-boost converter is a type of power electronic converter (switch-mode dc-dc converter)
- It comprises of a switch, a diode, an inductor and a capacitor
- Typically, the on-off switch is an IGBT transistor



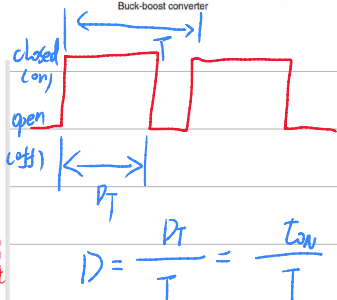
- When switch is CLOSED: energy supplied to inductor and I_L builds up
- When switch is OPEN: Inductor current flows through C, load, and diode, and I_L decreases
- As the change in inductor current during two states of the switch, we can use volt-sec balance of the



$$V_i [DT - V_o (1-D)] T = 0$$

$$V_o = \frac{D}{1-D} V_i$$

- The converter behaves as boost converter when $D \geq 0.5$.
- The converter behaves as buck converter when $D < 0.5$.



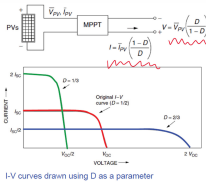
$$V = L \frac{di}{dt}$$

Example

Under certain ambient conditions, a PV module has its maximum power point at $V_m = 17$ volts and $I_m = 6$ A.

What duty cycle should an MPPT have if the module is delivering power to a $10\ \Omega$ resistance?

PV current-voltage curve with MPPT converter



Solution

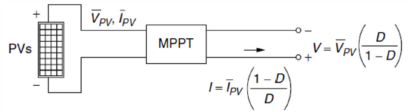
The maximum power delivered by the PVs is: $P = 17\text{ V} \times 6\text{ A} = 102\text{ W}$.

To deliver all of that 102 W to the $10\ \Omega$ resistor means the resistor needs a voltage V_R such that:

$$P = \frac{V_R^2}{R} = 102 = \frac{V_R^2}{10}$$
$$V_R = \sqrt{102 \times 10} = 31.9\text{ V}$$

The MPPT must bump the 17-V PV voltage to the desired 31.9V resistor voltage.

$$\frac{31.9}{17} = \left(\frac{D}{1-D}\right) = 1.88$$
$$D = \frac{1.88}{1+1.88} = 0.65$$



Q.19 Under certain ambient conditions, a PV module has its maximum power point at $V_m = 16$ V and $I_m = 4$ A. What duty cycle should an MPPT buck-boost converter have if the module is delivering the same power to a $1\ \Omega$ resistance?

$$P = V_m I_m = 64\text{ W}$$

$$P = \frac{V_R^2}{R} \Rightarrow V_R = 8\text{ V}$$

$$\frac{V_o}{V_i} = \frac{D}{1-D} \Rightarrow \frac{8}{16} = \frac{D}{1-D} \Rightarrow D = \frac{1}{3}$$

DC motor and battery lose about 15% of the available daily energy because they don't always operate on maximum power point

Estimating the system performance

- A good starting point to estimate system performance is the rated DC power output of an individual module under standard test conditions (STC)—that is, 1-sun, AM 1.5 and 25°C cell temperature. Then we can try to estimate the actual AC power output under varying conditions.

Actual ac power delivered at 1-sun,

$$P_{ac} = P_{dc,STC} \times (\text{Conversion Efficiency})$$

The conversion efficiency accounts for inverter efficiency, dirty collectors, mismatched modules, and differences in ambient conditions. Even in full sun, the impact of these losses can easily derate the power output by 20–40%.



Q.16 What are the various causes for the losses in output power in a PV system?

Example

Solution

Consider a PV array rated at 1 kW under standard test conditions. Module cell temperature is 53.8°C. DC power output at the MPP drops by 0.5%/°C above the STC temperature of 25°C.

With power loss at 0.5% per degree above 25°C, the DC rated power of the array would be:

$$P_{DC} = 1 \text{ kW} [1 - 0.005(53.8 - 25)] = 0.856 \text{ kW}$$

Estimate its ac output if there is a 3% array loss due to mismatched modules, line loss is 4%, and the inverter has an efficiency of 90%.

Including mismatch, dirt, and inverter efficiencies will result in an estimated AC rated power:

$$P_{AC} = 0.856 \text{ kW} \times 0.97 \times 0.96 \times 0.90 = 0.72 \text{ kW}$$

0.20 A PV panel under standard test conditions produces an output of 1000W. The output drops by 0.5%/°C due to temperature rise and 5% due to dirt accumulation on it. The ambient temperature is 45°C. Assuming power converter efficiency of 90%, determine the expected maximum AC power output from the panel.

Even though the system would be sold as a "1-kW system", the array will deliver only 72% of that as ac power to the load under the realistic conditions.

$$P_{DC} = 1 \text{ kW} [1 - 0.005(45 - 25)] = 900 \text{ W}$$

$$P_{AC} = P_{DC} \times 95\% \times 90\% = 769.5 \text{ W}$$

Capacity Factors for PV Grid-Connected Systems

Example: System sizing (first cut)

- If the system delivered full, rated power continuously, the capacity factor (CF) would be unity.

An energy efficient house in Singapore is to be fitted with a rooftop PV array that will annually displace all of the 1560 kWh/yr of electricity that the home uses. How many PV panels will be required and what area will be needed? Make assumptions as needed.

- Using data for Singapore, 4.1 kWh/m² day of annual insolation, using the peak hour approach.
- Energy (kWh/yr) = $P_{ac}(\text{kW}) \cdot (\text{h/day @ 1-sun}) \cdot 365 \text{ days/yr}$

$$P_{ac} = \frac{1560}{4.1 \times 365} = 1.0357 \text{ kW}$$

Considering the impacts of temperature, inverter efficiency, module mismatch, and dirt to come up with conversion efficiency of 75% from dc to ac.

$$P_{dc} = \frac{1.0357}{0.75} = 1.381 \text{ kW}$$

To estimate the collector efficiency, $P_{dc} = 1 \text{ kW/m}^2 \text{ insolation} \cdot A(\text{m}^2) \cdot \eta$

Assuming crystalline silicon modules which typically have an efficiency of about 12.5%, the area required

$$A = \frac{1.381 \text{ kW}}{1 \text{ kW/m}^2 \times 0.125} = 11.048 \text{ m}^2$$

0.21 If a small remote community requires 1MWh/month, determine the area of standard solar PV panels required. State the assumptions made in your calculation.

Using data for Singapore 4.1 kWh/m² - day of annual insolation, using the peak hour approach

$$\text{Energy (kWh/month)} = P_{ac}(\text{kW}) \cdot (\text{h/day @ 1-sun}) \cdot 30 \text{ days/month}$$

$$P_{ac} = \frac{1000 \text{ kWh}}{4.1 \times 30} = 8.13 \text{ kW}$$

assume DC → AC 75%

$$P_{dc} = \frac{P_{ac}}{75\%} = 10.84 \text{ kW}$$

To estimate the collector efficiency $P_{dc} = 1 \text{ kW/m}^2 \text{ insolation} \cdot A(\text{m}^2) \cdot \eta$

crystalline silicon modules have an efficiency of about 12.5%

$$A = \frac{10.84 \text{ kW}}{1 \text{ kW/m}^2 \times 12.5\%} = 86.72 \text{ m}^2$$

Number of modules required:

Module:	Sharp NE-K125U2	Kyocera KC158G	Shell SP150	Uni-Solar SSR256
Material:	Poly Crystal	Monocrystal	Monocrystal	Triple junction a-Si
Rated power $P_{dc,STC}$:	125 W	158 W	150 W	256 W
Voltage at max power:	26.0 V	23.2 V	34 V	66.0 V
Current at max power:	4.80 A	6.82 A	4.40 A	3.9
Open-circuit voltage V_{OC} :	32.3 V	28.9 V	43.4 V	95.2
Short-circuit current I_{SC} :	5.46 A	7.58 A	4.8 A	4.8
Length:	1.190 m	1.290 m	1.619 m	11.124 m
Width:	0.792 m	0.990 m	0.814 m	0.420 m
Efficiency:	13.3%	12.4%	11.4%	5.5%

Manufacturer:	Xantrex	Xantrex	Xantrex	Sunny Boy	Sunny Boy
Model:	STXR1500	STXR1500	PV 10	SB2000	SB2500
AC power:	1500 W	1500 W	10,000 W	2000 W	2500 W
AC voltage:	211–264 V	211–264 V	208 V, 3-φ	198–251 V	198–251 V
PV voltage range:	44–85 V	44–85 V	330–600 V	125–500 V	250–550 V
MPPT:					
Max input voltage:	120 V	120 V	600 V	500 V	600 V
Max input current:	—	—	31.9 A	10 A	11 A
Maximum efficiency:	92%	94%	95%	96%	94%

With three modules per string, rated voltage = $3 \times 23.2 = 69.6$ V, which is in the MPPT range.

This suggests using an array with three strings of three modules each, for a total of 9 modules

Roof area required = $9 \times 1.29 \times 0.99 = 11.49$ m² (very close to the initial estimate)

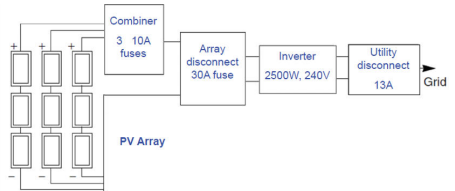
$$\text{Number of modules} = \frac{1381W}{158W/\text{module}} = 8.74$$

System sizing

- The short-circuit current for each string of panels is 7.58 A
- Applying the 125% factor to afford a safe oversizing margin, results in combiner fuses that must allow at least
Combiner fuses > $7.58 \text{ A} \times 1.25 = 9.475 \text{ A}$
- The array disconnect fuse must accommodate three such strings, so it must handle

$$\text{Array disconnect fuse} > 9.475 \times 3 = 28.425 \text{ A}$$

$$\text{Inverter fuse} > 1.25 \times \frac{2500 \text{ W}}{240 \text{ V}} = 13 \text{ A}$$



Module:	Sharp NE-K125U2	Kyocera KC158G	Shell SP150	Uni-Solar SSR256
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Length:	1.190 m	1.290 m	1.619 m	11.124 m
Width:	0.792 m	0.990 m	0.814 m	0.420 m
Efficiency:	13.3%	12.4%	11.4%	5.5%